Using common series

Key Points

Know the common series

Integers: $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$

Squares: $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$

Cubes: $\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4}$

To use these manipulate summations to contain these series

In addition: $\sum_{r=1}^{n} 1 = n$

Basic Skills

Q1. Using the common series find the following sums

a) $\sum_{r=1}^{5} r$ **b)** $\sum_{r=1}^{7} r^2$ **c)** $\sum_{r=1}^{11} r^3$ **d)** $1^2 + 2^2 + \cdots + 18^2$

e) $1^3 + 2^3 + \cdots 10^3$ **f)** $2\sum_{r=1}^{7} r^2 - \sum_{r=1}^{5} r$ **g)** $2^3 + 3^3 + \cdots + 20^3$

Q2. Work out the following $\sum_{r=5}^{10} r^2$

Q3. Show that the summation $\sum_{r=1}^{10}(r^2+2r+1)$ can written as $\sum_{r=1}^{10} r^2 + 2\sum_{r=1}^{10} r + \sum_{r=1}^{10} 1$ and hence work out the value

Q4. Work out the value of $\sum_{r=8}^{80} r^3$

Q5. Check that $\sum_{r=1}^{10}(r+1)^2 = \sum_{r=2}^{11} r^2$ and try to explain how we could arrive at this without explicit calculation

Competency Skills

Q6. Use the summation of an arithmetic series to prove that $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$

Q7. For the series $u_r = r(3r-1)$ find a formula for $\sum_{r=1}^{n} u_r$

Q8. **a)** Use sigma($\sum$) notation to write the series $(1\times1) + (2\times3) + (3\times5) + (4\times7) + \cdots + n(2n-1)$

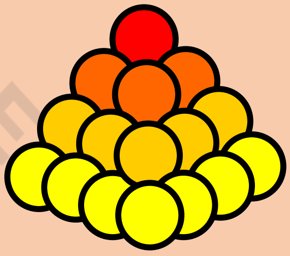
b) Hence find an expression for this series

Q9. Find the sum of the square numbers larger than n^2 but less than $9n^2$ in terms of n

Q10. Evaluate the series $(1\times2\times3) + (2\times3\times4) + (3\times4\times5) + \cdots + (18\times19\times20)$

Q11. Is it possible for the sum of the integers to equal 200

Advanced Skills



Q12. A square-based pyramid is made up of spheres, of radius r, stacked on top of one another with 100 spheres on the bottom layer.

a) Work out how many spheres are used in total

b) A similar square-based pyramid is made of squares but has the top cut off to form a frustum. The bottom layer has $4n^2$ spheres and the top layer has n^2, find the total number of spheres used in terms of n

Note: To consider the volume of the enclosed pyramid around the stack of spheres use the internal pyramid connected by centres and work outwards (Moderate geometry is needed)

Q13. To prove the sum of squares formula we can consider $(r-1)^3 = r^3 - 3r^2 + 3r - 1$ and summing both sides and rearranging to get $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$, complete this proof (do not use the cube formula)

Extension

Q14. Prove the formula for the sum of cubes and see if you can extend this in general to achieve Faulhaber's formula for the sum of any power.

Hint: Use $(r-1)^4$ with its binomial expansion and sum from 1 to n